

✓ **Fake and Real News project **

```
pip install -r requirements.txt
```

ERROR: Could not open requirements file: [Errno 2] No such file or directory: 'requirements.txt'

train_model.py

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
# Load datasets
fake = pd.read_csv("/content/Fake.csv")
true = pd.read_csv("/content/True.csv")
```

```
# Add labels
fake["label"] = 0
true["label"] = 1
```

```
# Combine datasets
data = pd.concat([fake, true], axis=0)
```

```
# Shuffle data
data = data.sample(frac=1, random_state=42)
```

```
# Features and target
X = data["text"]
y = data["label"]
```

```
# TF-IDF Vectorization
vectorizer = TfidfVectorizer(stop_words='english', max_df=0.7)

X = vectorizer.fit_transform(X)
```

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
# Train model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
▼ LogisticRegression ⓘ ?
LogisticRegression(max_iter=1000)
```

```
# Predictions
y_pred = model.predict(X_test)
```

```
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", round(accuracy * 100, 2), "%")
```

Accuracy: 98.22 %

```
import joblib
```

```
# Save model
joblib.dump(model, "model.pkl")
joblib.dump(vectorizer, "vectorizer.pkl")
```

```
print("Model Saved Successfully!")
```

Model Saved Successfully!

predict.py

```
import joblib
```

```
# Load model
model = joblib.load("model.pkl")
vectorizer = joblib.load("vectorizer.pkl")

while True:
    news = input("\nEnter News Text: ")

    if news.lower() == "exit":
        break

    news_vector = vectorizer.transform([news])

    prediction = model.predict(news_vector)

    if prediction[0] == 0:
        print("FAKE NEWS")
    else:
        print("REAL NEWS")
```

Enter News Text: The government announced a new education policy today.
FAKE NEWS

Enter News Text: Enter News Text: Scientists discovered a new planet similar to Earth.
FAKE NEWS

Enter News Text: The government approved the annual budget after parliamentary discussions.
REAL NEWS

Enter News Text: Scientists discovered a new species of fish in the Pacific Ocean.
FAKE NEWS

Enter News Text: The central bank announced no change in interest rates this month.
REAL NEWS

Enter News Text: